

9	B	Based on the definition of work done by spring tension.
10	A	Tension force on the sphere is always acting perpendicular to the displacement of the sphere and hence the work done on the sphere is zero. The electrical force on the sphere is in opposite direction to the electric field as the sphere is negatively charge and has a magnitude of qE and the displacement along the direction of the force is L. Work done by the electrical force = decrease in electrical potential energy, hence $U = -qEL$.
11	A	Free body diagram should not carry a centripetal force since it is not a real physical force